

DNMT analysis

Using the original orthofinder file in part 1.

PF00145 - 5mC PF01555 - 4mC

```
library(tidyverse)
```

```
library(ggtree)
file_5mC <- "data/genetree/dnmt_PF00145_candidates_plus_ref_domainonly.aln.clipkit.treefile"
file_4mC <- "data/genetree/dnmt_PF01555_candidates_plus_ref_domainonly.aln.clipkit.treefile"
tree_5mC <- ggtree::read.tree(file_5mC)
tree_4mC <- ggtree::read.tree(file_4mC)
```

Check if any tree_4mC\$tip.label overlap with tree_5mC\$tip.label.

```
overlap_tips <- intersect(tree_4mC$tip.label, tree_5mC$tip.label)
overlap_tips
```

```
## character(0)
```

```
# No overlap
```

Plot trees via daylight method

```
# metadata dd based on tree_5mC$tip.label and tree_4mC$tip.label split by /
dd_pre = data.frame(tip.label = c(tree_5mC$tip.label, tree_4mC$tip.label)) |>
  tidyr::separate(tip.label, into = c("species"), sep = "\\|", remove = FALSE) |>
  dplyr::mutate(
    # remove anything after CMT, DNMT, DRM and TRDMT
    species = stringr::str_remove(species, "CMT.*$"),
    species = stringr::str_remove(species, "DNM.*$"),
    species = stringr::str_remove(species, "DRM.*$"),
    species = stringr::str_remove(species, "TRDMT.*$"),
    species = stringr::str_remove(species, "DMT.*$")
  ) |>
  # if species == EctSp7 add gene name column for everything after / in tip label
  dplyr::mutate(
    gene_name = ifelse(grepl("EctSp7", tip.label),
      stringr::str_remove(tip.label, ".*\\|"),
      "")
  ) |>
  # Mp, Hs, At, Cn gets tip.label as gene_name
  dplyr::mutate(
    gene_name = dplyr::case_when(
      species %in% c("Mp", "Hs", "At", "Cn") ~ tip.label,
      TRUE ~ gene_name
    )
  ) |>
  dplyr::mutate(clade = dplyr::case_when(
    species %in% c("AscNod", "ChoLin", "DesDud", "DesHer", "DicDic",
      "EctCro", "EctFas", "EctSil", "EctSp7", "FucSer", "SacLat", "SacPo",
      "SchIsch", "TriMin", "ChrAus", "HetAka") ~ "early ochrophytina",
```

```

    species == "CyaPar" ~ "glaucophyta",
    species %in% c("PhaTri", "FisSol") ~ "diatoms",
    species %in% c("ChoCri", "CyaMer") ~ "red algae",
    species %in% c("Hs") ~ "animalia",
    species %in% c("At", "Mp") ~ "green lineage",
    TRUE ~ "others"
  )) |>
  dplyr::filter(
    !species %in% c("DisMes", "Cn") # remove DisMes and Cn due to low quality
    # !species %in% c("TriMin")
  )

deepclust_data2 <- readr::read_tsv("data/dnmt_PF00145_95_id_95_cov.txt",
                                   col_names = c("cluster_id", "sequence_id"))
deepclust_data2 |> dplyr::left_join(dd_pre, by = c("sequence_id" = "tip.label")) |>
  dplyr::mutate(cluster_size = dplyr::n(), .by = c(cluster_id, species)) |>
  dplyr::select(cluster_size, sequence_id, cluster_id, species) |>
  dplyr::filter(cluster_size > 1) |>
  # get sequences that are not cluster_id
  dplyr::filter(sequence_id != cluster_id) -> filt95.95

dd <- dd_pre |>
  dplyr::filter(
    !tip.label %in% filt95.95$sequence_id
  )

# Sisc", "Tribonema", "Chrysoparadoxa", "Heterosigma

# ggtree::ggtree(tree_5mC |> phytools::midpoint.root(), layout="daylight") %<+% dd +
#   geom_tiplab(aes(label = species), size=2)
# ggtree::ggtree(tree_4mC, layout="daylight") %<+% dd +
#   geom_tiplab(aes(label = species), size=2)

# # filter tip labels to only those in Orthogroup_filt.long
tree_5mC <-
  treeio::drop.tip(tree_5mC,
                   setdiff(tree_5mC$tip.label,
                           dd$tip.label))

# dd = as.data.frame(Orthogroup_filt.long |> dplyr::distinct(gene, gene_short, species, species_short,
# # set consistent colour for clade points and labels
#
colour_code <- c(
  "early ochrophytina" = "#D95F02",
  "phaeophyceae" = "#A6761D",
  "animalia" = "#7570b3",
  "red algae" = "darkred",
  "green lineage" = "#66a61e",
  "fungi" = "#e6ab02",
  "diatoms" = "gray40",
  "glaucophyta" ~ "#1b9e77",
  "others" = "black"
)

```

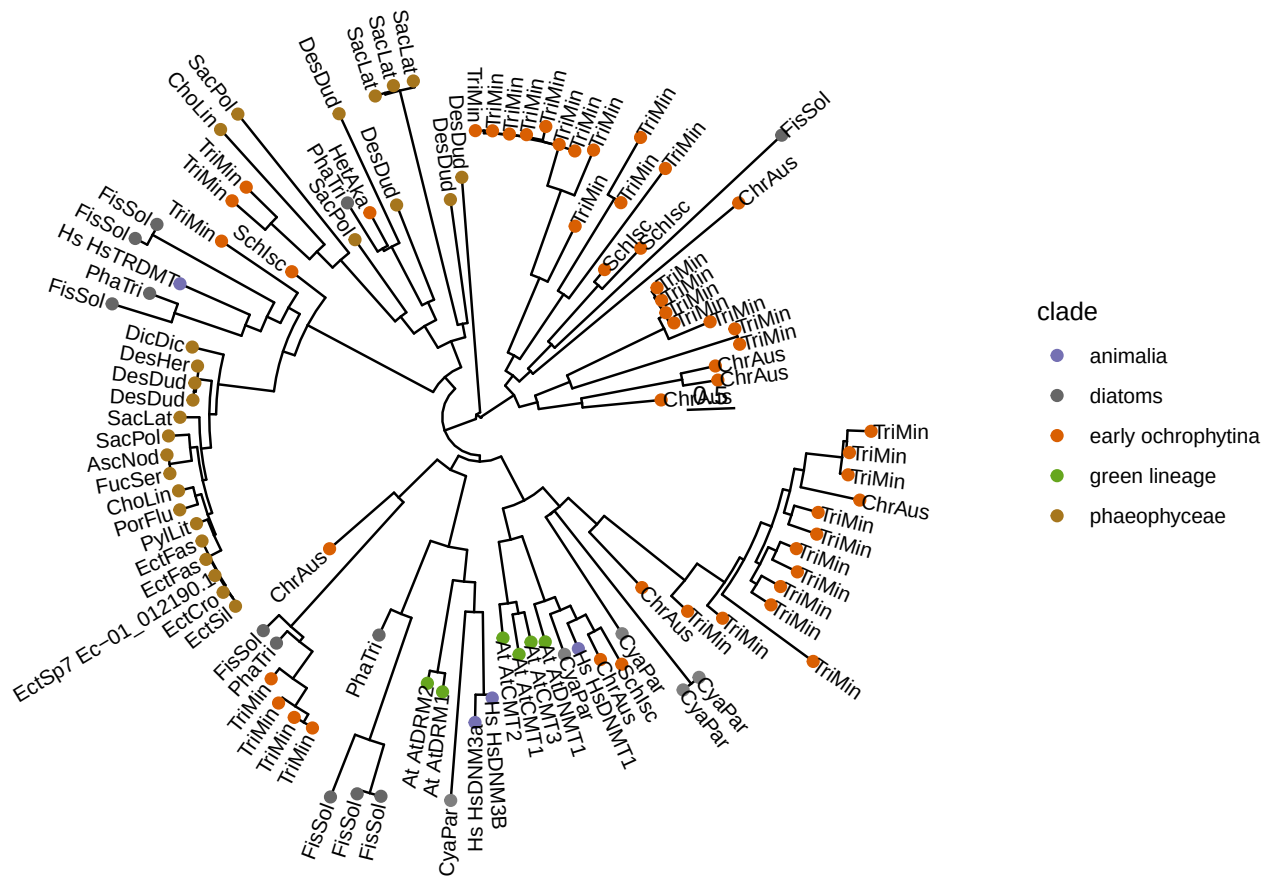
```

# colour by species_short
p1 <- ggtree::ggtree(tree_5mC |> phytools::midpoint.root(), layout="circular") %<+%
  dd +
  geom_tippoint(aes(colour = clade, label = clade),
    size=2) +
  geom_tiplab(aes(fill = clade,
    label = paste0(species, " ", gene_name)),
    # label = paste0(species_short, "\n", gene_short)),
    size=3,
    color = "black", # color for label font
    geom = "text",
    show.legend = FALSE) +
  ggplot2::scale_colour_manual(values = colour_code) +
  geom_treescale(width = 0.5)

# p2 <- ggtree::ggtree(tree_4mC |> phytools::midpoint.root(), layout="circular") %<+% dd +
#   geom_tippoint(aes(colour = clade, label = clade),
#     size=2) +
#   geom_tiplab(aes(fill = clade,
#     label = paste0(species, ":", gene_name)),
#     # label = paste0(species_short, "\n", gene_short)),
#     size=4,
#     color = "black", # color for label font
#     geom = "text",
#     show.legend = FALSE) +
#   ggplot2::scale_colour_manual(values = colour_code) +
#   geom_treescale(width = 0.5)

```

p1



```

    dplyr::distinct(protein_id, length)
Ec_domains <- domains_clean |>
  dplyr::filter(stringr::str_detect(protein_id, "Ec-01_012190.1")) |>
  dplyr::filter(db %in% c("PANTHER")) |>
  # dplyr::filter(db %in% c("Pfam", "Coils", "ProSitePatterns")) |>
  dplyr::mutate(
    xsubmin = start,
    xsubmax = end,
    subgene = description
  ) |>
  dplyr::full_join(Ec_gene_info, by = c("protein_id", "length")) |>
  # dplyr::left_join(Orthogroup_filt.long |>
  #   dplyr::select(gene, Orthogroup),
  #   by = c("protein_id" = "gene")) |>
  dplyr::mutate(protein_id_short = stringr::str_replace(protein_id, "\\.[0-9]+$", ""))

# add "Histone methylation protein DOT1" to start 1195 and end 1253 for Ec-25_002640.1 for Ec_domains

# Base gene features (full-length proteins)
p <- Ec_domains |>
  dplyr::distinct(protein_id_short, length) |>
  ggplot2::ggplot(
    ggplot2::aes(
      xmin = 1,
      xmax = length,
      y = protein_id_short
    )
  ) +
  gggenes::geom_gene_arrow(
    arrowhead_height = grid::unit(0, "mm"),
    arrowhead_width = grid::unit(0, "mm"),
    fill = "grey90",
    colour = "black"
  ) +
  gggenes::theme_genes() +
  ggplot2::theme_void() +
  ggplot2::labs(
    title = "Ec DOT1 domains from InterProScan"
  )

# Add domains as subgenes
p +
  gggenes::geom_subgene_arrow(
    data = Ec_domains,
    ggplot2::aes(
      xsubmin = xsubmin,
      xsubmax = xsubmax,
      fill = subgene
    ),
    arrowhead_height = grid::unit(0, "mm"),
    arrowhead_width = grid::unit(0, "mm"),
    colour = "black"
  ) +
  ggplot2::theme(

```

```

        legend.position = "bottom",
        # add gene labels
        axis.text.y = ggplot2::element_text(size = 8)
      ) +
    ggplot2::guides(
      fill = ggplot2::guide_legend(
        title = "Domains",
        nrow = 2,
        byrow = TRUE
      )
    ) +
    # use a colour bind friendly palette from paletteer
    paletteer::scale_fill_paletteer_d("nbapalettes::supersonics_holiday")

```

Ec DOT1 domains from InterProScan

Ec-01_012190



Domains



TRNA (CYTOSINE(38)-C(5))-METHYLTRANSFERASE

```
# ggplot2::ggsave("figure/Ec_DOT1_domain_structure.pdf_domain_structure_v2.pdf", width = 4, height = 2)
```

```
devtools::session_info(pkgs = "attached")
```

```
## - Session info -----
```

```
## setting value
```

```
## version R version 4.4.0 (2024-04-24)
```

```
## os macOS 15.7.2
```

```
## system x86_64, darwin20
```

```
## ui X11
```

```
## language (EN)
```

```
## collate en_US.UTF-8
```

```
## ctype en_US.UTF-8
```

```
## tz Europe/Berlin
```

```
## date 2026-01-30
```

```
## pandoc 3.4 @ /Applications/RStudio.app/Contents/Resources/app/quarto/bin/tools/x86_64/ (via rmark
```

```
##
```

```
## - Packages -----
## package * version date (UTC) lib source
## dplyr * 1.1.4 2023-11-17 [1] CRAN (R 4.4.0)
## forcats * 1.0.0 2023-01-29 [1] CRAN (R 4.4.0)
## gggenes * 0.5.1 2023-09-05 [1] CRAN (R 4.4.0)
## ggplot2 * 4.0.0 2025-09-11 [1] CRAN (R 4.4.1)
## ggtree * 4.1.1 2025-11-17 [1] Github (YuLab-SMU/ggtree@57a3e6b)
## lubridate * 1.9.4 2024-12-08 [1] CRAN (R 4.4.1)
## purrr * 1.1.0 2025-07-10 [1] CRAN (R 4.4.1)
## readr * 2.1.5 2024-01-10 [1] CRAN (R 4.4.0)
## stringr * 1.5.2 2025-09-08 [1] CRAN (R 4.4.1)
## tibble * 3.3.0 2025-06-08 [1] CRAN (R 4.4.1)
## tidyr * 1.3.1 2024-01-24 [1] CRAN (R 4.4.0)
## tidyverse * 2.0.0 2023-02-22 [1] CRAN (R 4.4.0)
##
## [1] /Library/Frameworks/R.framework/Versions/4.4-x86_64/Resources/library
##
## -----
```